

```
In [1]: import pandas as pd
        from sklearn import preprocessing
```

```
In [2]: df=pd.read_csv(r"C:\Users\Dell\Uber Request Data.csv")
```

```
In [3]: df
```

```
Out[3]:
```

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp
0	619	Airport	1.0	Trip Completed	11/7/2016 11:51	11/7/2016 13:00
1	867	Airport	1.0	Trip Completed	11/7/2016 17:57	11/7/2016 18:47
2	1807	City	1.0	Trip Completed	12/7/2016 9:17	12/7/2016 9:58
3	2532	Airport	1.0	Trip Completed	12/7/2016 21:08	12/7/2016 22:03
4	3112	City	1.0	Trip Completed	13-07-2016 08:33:16	13-07-2016 09:25:47
...	...	...	...	...	...	...
6740	6745	City	NaN	No Cars Available	15-07-2016 23:49:03	NaN
6741	6752	Airport	NaN	No Cars Available	15-07-2016 23:50:05	NaN
6742	6751	City	NaN	No Cars Available	15-07-2016 23:52:06	NaN
6743	6754	City	NaN	No Cars Available	15-07-2016 23:54:39	NaN
6744	6753	Airport	NaN	No Cars Available	15-07-2016 23:55:03	NaN

6745 rows × 6 columns

```
In [4]: df.head()
```

Out[4]:	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp
0	619	Airport	1.0	Trip Completed	11/7/2016 11:51	11/7/2016 13:00
1	867	Airport	1.0	Trip Completed	11/7/2016 17:57	11/7/2016 18:47
2	1807	City	1.0	Trip Completed	12/7/2016 9:17	12/7/2016 9:58
3	2532	Airport	1.0	Trip Completed	12/7/2016 21:08	12/7/2016 22:03
4	3112	City	1.0	Trip Completed	13-07-2016 08:33:16	13-07-2016 09:25:47

In [5]: `df.tail()`

Out[5]:	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp
6740	6745	City	NaN	No Cars Available	15-07-2016 23:49:03	NaN
6741	6752	Airport	NaN	No Cars Available	15-07-2016 23:50:05	NaN
6742	6751	City	NaN	No Cars Available	15-07-2016 23:52:06	NaN
6743	6754	City	NaN	No Cars Available	15-07-2016 23:54:39	NaN
6744	6753	Airport	NaN	No Cars Available	15-07-2016 23:55:03	NaN

In [6]: `df.shape`

Out[6]: (6745, 6)

In [7]: `df.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6745 entries, 0 to 6744
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Request id            6745 non-null   int64
1   Pickup point          6745 non-null   object
2   Driver id             4095 non-null   float64
3   Status                6745 non-null   object
4   Request timestamp     6745 non-null   object
5   Drop timestamp        2831 non-null   object
dtypes: float64(1), int64(1), object(4)
memory usage: 316.3+ KB

```

In [8]: `df.describe()`

```

Out[8]:

```

	Request id	Driver id
count	6745.000000	4095.000000
mean	3384.644922	149.501343
std	1955.099667	86.051994
min	1.000000	1.000000
25%	1691.000000	75.000000
50%	3387.000000	149.000000
75%	5080.000000	224.000000
max	6766.000000	300.000000

In [9]: `df.dtypes`

```

Out[9]:
Request id            int64
Pickup point          object
Driver id             float64
Status                object
Request timestamp     object
Drop timestamp        object
dtype: object

```

In [11]: `df.isnull()`

Out[11]:

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
...	...	...	...	...	...	...
6740	False	False	True	False	False	True
6741	False	False	True	False	False	True
6742	False	False	True	False	False	True
6743	False	False	True	False	False	True
6744	False	False	True	False	False	True

6745 rows × 6 columns

In [12]: `df.notnull()`

Out[12]:

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp
0	True	True	True	True	True	True
1	True	True	True	True	True	True
2	True	True	True	True	True	True
3	True	True	True	True	True	True
4	True	True	True	True	True	True
...	...	...	...	...	...	...
6740	True	True	False	True	True	False
6741	True	True	False	True	True	False
6742	True	True	False	True	True	False
6743	True	True	False	True	True	False
6744	True	True	False	True	True	False

6745 rows × 6 columns

In [13]: `df['Request timestamp']=pd.to_datetime(df['Request timestamp'],format='mixed',dayfi`

In [14]: `df.dtypes`

```
Out[14]: Request id          int64
Pickup point          object
Driver id             float64
Status                object
Request timestamp     datetime64[ns]
Drop timestamp        object
dtype: object
```

```
In [15]: df.head()
```

```
Out[15]:
```

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp
0	619	Airport	1.0	Trip Completed	2016-07-11 11:51:00	11/7/2016 13:00
1	867	Airport	1.0	Trip Completed	2016-07-11 17:57:00	11/7/2016 18:47
2	1807	City	1.0	Trip Completed	2016-07-12 09:17:00	12/7/2016 9:58
3	2532	Airport	1.0	Trip Completed	2016-07-12 21:08:00	12/7/2016 22:03
4	3112	City	1.0	Trip Completed	2016-07-13 08:33:16	13-07-2016 09:25:47

```
In [17]: df['Pickup point']=df['Pickup point'].map({'Airport':0,'City':1})
```

```
In [18]: df
```

Out[18]:

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp
0	619	0	1.0	Trip Completed	2016-07-11 11:51:00	11/7/2016 13:00
1	867	0	1.0	Trip Completed	2016-07-11 17:57:00	11/7/2016 18:47
2	1807	1	1.0	Trip Completed	2016-07-12 09:17:00	12/7/2016 9:58
3	2532	0	1.0	Trip Completed	2016-07-12 21:08:00	12/7/2016 22:03
4	3112	1	1.0	Trip Completed	2016-07-13 08:33:16	13-07-2016 09:25:47
...	...	...	...	...	...	...
6740	6745	1	NaN	No Cars Available	2016-07-15 23:49:03	NaN
6741	6752	0	NaN	No Cars Available	2016-07-15 23:50:05	NaN
6742	6751	1	NaN	No Cars Available	2016-07-15 23:52:06	NaN
6743	6754	1	NaN	No Cars Available	2016-07-15 23:54:39	NaN
6744	6753	0	NaN	No Cars Available	2016-07-15 23:55:03	NaN

6745 rows × 6 columns

In [19]:

df

Out[19]:

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp
0	619	0	1.0	Trip Completed	2016-07-11 11:51:00	11/7/2016 13:00
1	867	0	1.0	Trip Completed	2016-07-11 17:57:00	11/7/2016 18:47
2	1807	1	1.0	Trip Completed	2016-07-12 09:17:00	12/7/2016 9:58
3	2532	0	1.0	Trip Completed	2016-07-12 21:08:00	12/7/2016 22:03
4	3112	1	1.0	Trip Completed	2016-07-13 08:33:16	13-07-2016 09:25:47
...	...	...	...	...	...	...
6740	6745	1	NaN	No Cars Available	2016-07-15 23:49:03	NaN
6741	6752	0	NaN	No Cars Available	2016-07-15 23:50:05	NaN
6742	6751	1	NaN	No Cars Available	2016-07-15 23:52:06	NaN
6743	6754	1	NaN	No Cars Available	2016-07-15 23:54:39	NaN
6744	6753	0	NaN	No Cars Available	2016-07-15 23:55:03	NaN

6745 rows × 6 columns

In [20]: `df.isnull().sum()`

```
Out[20]: Request id      0
Pickup point    0
Driver id      2650
Status          0
Request timestamp  0
Drop timestamp   3914
dtype: int64
```

In [21]: `df['Driver id']=df['Driver id'].fillna(df['Driver id'].mean())`

In [22]: `df.isnull().sum()`

```
Out[22]: Request id      0
Pickup point    0
Driver id       0
Status          0
Request timestamp 0
Drop timestamp   3914
dtype: int64
```

```
In [26]: from sklearn.preprocessing import MinMaxScaler
scaler=MinMaxScaler()
df[['Driver id','Request id']]=scaler.fit_transform(df[['Driver id','Request id']])
df
```

```
Out[26]:
```

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp
0	0.091353	0	0.00000	Trip Completed	2016-07-11 11:51:00	11/7/2016 13:00
1	0.128012	0	0.00000	Trip Completed	2016-07-11 17:57:00	11/7/2016 18:47
2	0.266962	1	0.00000	Trip Completed	2016-07-12 09:17:00	12/7/2016 9:58
3	0.374132	0	0.00000	Trip Completed	2016-07-12 21:08:00	12/7/2016 22:03
4	0.459867	1	0.00000	Trip Completed	2016-07-13 08:33:16	13-07-2016 09:25:47
...	...	...	...	...	...	...
6740	0.996896	1	0.49666	No Cars Available	2016-07-15 23:49:03	NaN
6741	0.997931	0	0.49666	No Cars Available	2016-07-15 23:50:05	NaN
6742	0.997783	1	0.49666	No Cars Available	2016-07-15 23:52:06	NaN
6743	0.998226	1	0.49666	No Cars Available	2016-07-15 23:54:39	NaN
6744	0.998078	0	0.49666	No Cars Available	2016-07-15 23:55:03	NaN

6745 rows × 6 columns

```
In [ ]:
```